FILTERED COLIMITS OF K-FLAT COMPLEXES ARE K-FLAT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Tensor products commute with colimits, so the total tensor product commutes with termwise colimits. Algebra, Lemma \ref{algebra-lemma-tensor-products-commute-with-limits}, cited at proof lines 14325-14327. Identifies the tensor product against the colimit with the colimit of tensor products.
- Filtered colimits of modules are exact. Algebra, Lemma \ref{algebra-lemma-directed-colimit-exact}, cited at proof lines 14330-14331. Transfers the quasi-isomorphisms through the colimit.

Source reference. Original lemma 06Y3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8872–8872. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00DD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 493–493. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00DB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 491–491. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $K_1^\bullet \to K_2^\bullet \to \ldots$ be a system of K-flat complexes. Then $\mathrm{colim}_i K_i^\bullet$ is K-flat. More generally any filtered colimit of K-flat complexes is K-flat.*

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Proof. Because we are taking termwise colimits we have

$$\operatorname{colim}_i \operatorname{Tot}(M^\bullet \otimes_R K_i^\bullet) = \operatorname{Tot}(M^\bullet \otimes_R \operatorname{colim}_i K_i^\bullet)$$

by Algebra, Lemma 00DD. Hence the lemma follows from the fact that filtered colimits are exact, see Algebra, Lemma 00DB. □